Sukwon Yun

Email • Homepage • Google Scholar • Github

RESEARCH INTEREST

AI for Science

Building AI systems that collaborate, learn from experiments, and close the loop in scientific discovery

- Methods: AI Co-Scientists, Multi-Agent Systems, Mixture-of-Experts, Multimodal Learning
- Domains: Virtual Cell, Protein Design

EDUCATION

University of North Carolina at Chapel Hill, Chapel Hill, NC, USA — Aug 2024 – Dec 2025
- M.S. in Computer Science (Advisor: Prof. Tianlong Chen)

KAIST, Daejeon, South Korea — Aug 2021 – Aug 2023
- M.S. in Industrial and Systems Engineering (Advisor: Prof. Chanyoung Park)

Hanyang University, Seoul, South Korea — Mar 2015 – Aug 2021
- B.S. in Industrial Engineering, *Summa Cum Laude* (incl. 2-year military service)

POSITIONS & RESEARCH EXPERIENCE

Xaira Therapeutics, South San Francisco, CA, USA
- AI Scientist — Sep 2026 – Present
  - Manager: Prof. Bo Wang
  - Test-time scaling and Post-training for De Novo Antibody Design
- AI Resident — Jun 2026 – Sep 2026
  - Mentors: Prof. Bo Wang, Dr. Mehran Karimzadeh, Dr. Yongju Lee
  - Project: Autoresearch for Virtual Cell

Genentech, South San Francisco, CA, USA — May 2025 – Aug 2025
- Research Intern
  - Mentors: Prof. Aviv Regev, Dr. Russell Littman
  - Project: Mixture-of-Experts for Vision Transformers (*NeurIPS 2025 AI for Science*)

Tokyo Institute of Technology (Tokyo Tech), Tokyo, Japan — Oct 2022 – Feb 2023
- Visiting Researcher
  - Host: Prof. Tsuyoshi Murata
  - Project: Graph Diffusion for Missing Feature Problem (*KDD 2025*)

SELECTED PUBLICATIONS
(FULL LIST IN THE END)
*: Equal contribution

Graph-of-Agents: A Graph-Based Framework for Multi-Agent LLM Collaboration
Sukwon Yun, *et al.*, Tianlong Chen
International Conference on Learning Representations (**ICLR 2026**)

Skill-Based Mixture-of-Experts: Adaptive Routing for Heterogeneous Reasoning via Inferred Skills
Justin CY Chen*, **Sukwon Yun***, *et al.*, Mohit Bansal
International Conference on Machine Learning (**ICML 2026**)

Mew: Multiplexed Immunofluorescence Image Analysis through an Efficient Multiplex Network
Sukwon Yun, *et al.*, Tianlong Chen
European Conference on Computer Vision (**ECCV 2024**)

Flex-MoE: Modeling Arbitrary Modality Combination via the Flexible Mixture-of-Experts
Sukwon Yun, *et al.*, Tianlong Chen
Conference on Neural Information Processing Systems (**NeurIPS 2024 Spotlight**)

When to Trust Your AI Co-Scientist? Learning Prior Reliability in Closed-Loop Bayesian Optimization
Sukwon Yun, *et al.*, Su-In Lee
Machine Learning in Computational Biology (**MLCB 2026 Spotlight**)

Single-cell RNA-seq data imputation using Feature Propagation
Sukwon Yun*, *et al.*, Chanyoung Park
International Conference on Machine Learning (**ICML 2023 Workshop on Comp. Bio. Best Paper Award**)

AWARDS & SCHOLARSHIPS

NeurIPS Top Reviewer 2025
Best Paper Award, AAAI Workshop on Large Language Models and Generative AI for Health 2025
Best Paper Award, KDD Workshop on Federated Learning for Data Mining and Graph Analytics 2024
Best Paper Award, ICML Workshop on Computational Biology 2023

Korea National Scholarship 2021 – 2023
- Awarded by the Ministry of Science and ICT, South Korea

Poster Competition Excellence Award 2022
- Awarded at Industrial/Social Problem Solving Session held by Department of ISysE, KAIST

SIGIR Student Travel Award 2022
- ACM International Conference on Information and Knowledge Management, Georgia, USA

Merit Based Scholarship 2021
- Awarded by the Department of Industrial Engineering, Hanyang University

Hanyang Academic Achievement Award 2021
- Awarded within the top 3% among the College of Engineering, Hanyang University

Certificate of Recognition 2018
- Awarded when serving military service as an auxiliary police by Seoul Metropolitan Police

TEACHING EXPERIENCE

University of North Carolina at Chapel Hill
- COMP283: Discrete Structures Fall, 2025

KAIST
- IE343: Statistical Machine Learning Spring, 2022

PROFESSIONAL SERVICES

Event Organizations
- Session Chair, Knowledge Representation 3, Birmingham, UK, CIKM (2023)

Program Committee
- AAAI (2025, 2026)

Conference Reviews
- NeurIPS (2023 -), ICLR (2024 -), ICML (2024 -), CVPR (2025), ICCV (2025), ECCV (2024), MICCAI (2025), ACL (2025), COLM (2025), KDD (2025), CPAL (2024 -)

Workshop Reviews
- ICBINB-BIO @ NeurIPS (2026), GenAI4Health @ AAAI (2025), ML4LMS @ ICML (2024)

FULL PUBLICATIONS

*: Equal contribution
#: Co-correspondence

CONFERENCES

[C24] When to Trust Your AI Co-Scientist? Learning Prior Reliability in Closed-Loop Bayesian Optimization
Sukwon Yun, Xiaojian Chen, Su-In Lee
Machine Learning in Computational Biology (**MLCB 2026 Spotlight**)

[C23] SPATIA: Multimodal Generation and Prediction of Spatial Cell Phenotypes
Zhenglun Kong, Mufan Qiu, John Boesen, Xiang Lin, **Sukwon Yun**, Tianlong Chen, Manolis Kellis, Marinka Zitnik
International Conference on Machine Learning (**ICML 2026**)

[C22] Skill-Based Mixture-of-Experts: Adaptive Routing for Heterogeneous Reasoning via Inferred Skills
Justin Chih-Yao Chen*, **Sukwon Yun***, Elias Stengel-Eskin*, Tianlong Chen, Mohit Bansal
International Conference on Machine Learning (**ICML 2026**)

[C21] Graph-of-Agents: A Graph-Based Framework for Multi-Agent LLM Collaboration
Sukwon Yun, Jie Peng, Pingzhi Li, Wendong Fan, Jie Chen, James Zou, Guohao Li, Tianlong Chen
International Conference on Learning Representations (**ICLR 2026**)

[C20] BetaConform: Efficient MAP Estimation of LLM Ensemble Judgment Performance with Prior Transfer
Huaizhi Qu, Inyoung Choi, Zhen Tan, Song Wang, **Sukwon Yun**, Qi Long, Faizan Siddiqui, Kwonjoon Lee, Tianlong Chen
Conference on Neural Information Processing Systems (**NeurIPS 2025**)

[C19] Training Robust Graph Neural Networks by Modeling Noise Dependencies
Yeonjun In, Kanghoon Yoon, **Sukwon Yun**, Kibum Kim, Sungchul Kim, Chanyoung Park
Conference on Neural Information Processing Systems (**NeurIPS 2025**)

[C18] Oldie but Goodie: Re-illuminating Label Propagation on Graphs with Partially Observed Features
Sukwon Yun, Xin Liu, Yunhak Oh, Junseok Lee, Tianlong Chen, Tsuyoshi Murata, Chanyoung Park
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2025**)

[C17] Spatial Coordinates as a Cell Language: A Multi-Sentence Framework for Imaging Mass Cytometry Analysis
Chi-Jane Chen*, Yuhang Chen*, **Sukwon Yun***, Natalie Stanley, Tianlong Chen
Annual Meeting of the Association for Computational Linguistics (**ACL 2025 Findings**)

[C16] GRNFormer: A Biologically-Guided Framework for Integrating Gene Regulatory Networks into RNA Foundation Models
Mufan Qiu, Xinyu Hu, Fengwei Zhan, **Sukwon Yun**, Jie Peng, Ruichen Zhang, Bhavya Kailkhura, Jiekun Yang, Tianlong Chen
Annual Meeting of the Association for Computational Linguistics (**ACL 2025 Findings**)

[C15] Agents Under Siege: Breaking Pragmatic Multi-Agent LLM Systems with Optimized Prompt Attacks
Rana Shahroz, Zhen Tan, **Sukwon Yun**, Charles Fleming, Tianlong Chen
Annual Meeting of the Association for Computational Linguistics (**ACL 2025 Oral**)

[C14] I^2MoE: Interpretable Multimodal Interaction-aware Mixture-of-Experts
Jiayi Xin, **Sukwon Yun**, Jie Peng, Inyoung Choi, Jenna L. Ballard, Tianlong Chen, Qi Long
International Conference on Machine Learning (**ICML 2025**)

[C13] Modalities Contribute Unequally: Enhancing Medical Multi-modal Learning through Adaptive Modality Token Re-balancing
Jie Peng, Jenna L. Ballard, Mohan Zhang, **Sukwon Yun**, Jiayi Xin, Qi Long, Yanyong Zhang, Tianlong Chen
International Conference on Machine Learning (**ICML 2025**)

[C12] Sparse MoE as a New Treatment: Addressing Forgetting, Fitting, Learning Issues in Multi-Modal Multi-Task Learning
Jie Peng, **Sukwon Yun**, Kaixiong Zhou, Ruida Zhou, Thomas Hartvigsen, Yanyong Zhang, Zhangyang Wang, Tianlong Chen
Conference on Parsimony and Learning (**CPAL 2025**)

[C11] Cut the Crap: An Economical Communication Pipeline for LLM-based Multi-Agent Systems
Guibin Zhang, Yanwei Yue, Zhixun Li, **Sukwon Yun**, Guancheng Wan, Kun Wang, Dawei Cheng, Jeffrey Xu Yu, Tianlong Chen
International Conference on Learning Representations (**ICLR 2025**)

[C10] PortLLM: Personalizing Evolving Large Language Models with Training-Free and Portable Model Patches
Rana Muhammad Shahroz Khan, Pingzhi Li*, **Sukwon Yun***, Zhenyu Wang, Shahriar Nirjon, Chau-Wai Wong, Tianlong Chen
International Conference on Learning Representations (**ICLR 2025**)

[C9] Subgraph Federated Learning for Local Generalization
Sungwon Kim, Yoonho Lee, Yunhak Oh, Namkyeong Lee, **Sukwon Yun**, Junseok Lee, Sein Kim, Carl Yang, Chanyoung Park
International Conference on Learning Representations (**ICLR 2025 Oral**)

[C8] Flex-MoE: Modeling Arbitrary Modality Combination via the Flexible Mixture-of-Experts
Sukwon Yun, Inyoung Choi, Jie Peng, Yangfan Wu, Jingxuan Bao, Qiyiwen Zhang, Jiayi Xin, Qi Long[#], Tianlong Chen[#]
Conference on Neural Information Processing Systems (**NeurIPS 2024 Spotlight**)

[C7] DALK: Dynamic Co-Augmentation of LLMs and KG to answer Alzheimer's Disease Questions with Scientific Literature
Dawei Li, Shu Yang, Zhen Tan, Jae Young Baik, **Sukwon Yun**, Joseph Lee, Aaron Chacko, Bojian Hou, Duy Duong-Tran, Ying Ding, Huan Liu, Li Shen, Tianlong Chen
Conference on Empirical Methods in Natural Language Processing (**EMNLP 2024 Findings**)

[C6] Mew: Multiplexed Immunofluorescence Image Analysis through an Efficient Multiplex Network
Sukwon Yun, Jie Peng, Alexandro Trevino, Chanyoung Park, Tianlong Chen
European Conference on Computer Vision (**ECCV 2024**)

[C5] DEGNN: Dual Experts Graph Neural Network Handling Both Edge and Node Feature Noise
Tai Hasegawa, **Sukwon Yun**, Xin Liu, Yin Jun Phua, Tsuyoshi Murata
Pacific-Asia Conference on Knowledge Discovery and Data Mining (**PAKDD 2024**)

[C4] MUSE: Music Recommender System with Shuffle Play Recommendation Enhancement
Yunhak Oh*, **Sukwon Yun***, Dongmin Hyun, Sein Kim, Chanyoung Park
ACM International Conference on Information and Knowledge Management (**CIKM 2023**)

[C3] S-Mixup: Structural Mixup for Graph Neural Networks
Junghurn Kim*, **Sukwon Yun***, Chanyoung Park
ACM International Conference on Information and Knowledge Management (**CIKM 2023 Short**)

[C2] MELT: Mutual Enhancement of Long-Tailed User and Item for Sequential Recommendation
Kibum Kim, Dongmin Hyun, **Sukwon Yun**, Chanyoung Park
ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2023**)

[C1] LTE4G: Long-Tail Experts for Graph Neural Networks
Sukwon Yun, Kibum Kim, Kanghoon Yoon, Chanyoung Park
ACM International Conference on Information and Knowledge Management (**CIKM 2022**)

JOURNALS

[J2] Single-cell multiregion epigenomic rewiring in Alzheimer's disease progression and cognitive resilience
Zunpeng Liu*, Shanshan Zhang*, Benjamin T. James, Kyriaki Galani, Riley J. Mangan, Stuart Benjamin Fass, Chuqian Liang, Manoj M Wagle, Carles A. Boix, Yosuke Tanigawa, **Sukwon Yun**, Yena Sung, Xushen Xiong, Na Sun, Lei Hou, Martin Wohlwend, Mufan Qiu, Xikun Han, Lei Xiong, Efthalia Preka, Lei Huang, William F. Li, Li-Lun Ho, Amy Grayson, Julio Mantero, Alexey Kozlenkov, Hansruedi Mathys, Tianlong Chen, Stella Dracheva, David A. Bennett, Li-Huei Tsai, Manolis Kellis
Cell 2025

[J1] Single-cell RNA sequencing data imputation using bi-level feature propagation
Junseok Lee*, **Sukwon Yun***, Yeongmin Kim, Tianlong Chen, Manolis Kellis, Chanyoung Park
Briefings in Bioinformatics 2024

WORKSHOPS

[W7] Sparse Mixture-of-Experts for Multi-Channel Imaging: Are All Channel Interactions Required?
Sukwon Yun, Heming Yao, Burkhard Hoeckendorf, David Richmond, Aviv Regev[#], Russell Littman[#]
NeurIPS 2025 AI for Science

[W6] Graph-of-Agents: A Graph-Based Framework for Multi-Agent LLM Collaboration
Sukwon Yun, Jie Peng, Pingzhi Li, Wendong Fan, Jie Chen, James Zou, Guohao Li, Tianlong Chen
ICML 2025 MAS

[W5] One Leaf Knows Autumn: A Piece of Data-Model Facilitates Efficient Cancer Prognosis with Histological and Genomic Modalities
Jie Peng, Jingxia Jiang, Yueliang Ying, **Sukwon Yun**, Qi Long, Yanyong Zhang, Tianlong Chen
AAAI GenAI4Health 2025

[W4] Generate, Then Retrieve: Addressing Missing Modalities in Multimodal Learning via Generative AI and MoE
Sukwon Yun*, Jiayi Xin*, Inyoung Choi, Jie Peng, Qi Long, Tianlong Chen
AAAI GenAI4Health 2025 Best Paper Award

[W3] Subgraph Federated Learning for Local Generalization
Sungwon Kim, Yoonho Lee, Yunhak Oh, Namkyeong Lee, **Sukwon Yun**, Junseok Lee, Sein Kim, Carl Yang, Chanyoung Park
KDD 2024 FedKDD Best Paper Award

[W2] Noise Robust Graph Learning under Feature-Dependent Graph-Noise
Yeonjun In, Kanghoon Yoon, **Sukwon Yun**, Kibum Kim, Sungchul Kim, Chanyoung Park
WWW 2024 DCAI Oral

[W1] Single-cell RNA-seq data imputation using Feature Propagation
Sukwon Yun*, Junseok Lee*, Chanyoung Park
ICML 2023 WCB Best Paper Award